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# Introduction to Probability

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# Notation

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| $v$ | velocity |

# Preface

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# Probability

# Chapter 1

# Basic Concepts

## 1.1 Probability Distributions

**Definition 1.1.0.1.** Let $\Omega$ be a set and $\mathcal{P} \subset \mathcal{P}(X)$. Then $\mathcal{P}$ is said to be a **$\pi$-system** on $\Omega$ if for each $A, B \in \mathcal{P}$, $A \cap B \in \mathcal{P}$.

**Definition 1.1.0.2.** Let $\Omega$ be a set and $\mathcal{L} \subset \mathcal{P}(\Omega)$. Then $\mathcal{L}$ is said to be a **$\lambda$-system** on $\Omega$ if

1. $\mathcal{L} \neq \varnothing$
2. for each $A \in \mathcal{L}$, $A^c \in \mathcal{L}$
3. for each $(A_n)_{n\in\mathbb{N}} \subset \mathcal{L}$, if $(A_n)_{n\in\mathbb{N}}$ is disjoint, then $\bigcup\limits_{n\in\mathbb{N}} A_n \in \mathcal{L}$

**Exercise 1.1.0.3.** Let $\Omega$ be a set and $\mathcal{L}$ a $\lambda$-system on $\Omega$. Then

1. $\Omega, \varnothing \in \mathcal{L}$

*Proof.* Straightforward. □

**Definition 1.1.0.4.** Let $\Omega$ be a set and $\mathcal{C} \subset \mathcal{P}(\Omega)$. Put

$$\mathcal{S} = \{\mathcal{L} \subset \mathcal{P}(\Omega) : \mathcal{L} \text{ is a } \lambda\text{-system on } \Omega \text{ and } \mathcal{C} \subset \mathcal{L}\}$$

We define the **$\lambda$-system on $\Omega$ generated by** $\mathcal{C}$, $\lambda(\mathcal{C})$, to be

$$\lambda(\mathcal{C}) = \bigcap_{\mathcal{L}\in\mathcal{S}} \mathcal{L}$$

**Exercise 1.1.0.5.** Let $\Omega$ be a set and $\mathcal{C} \subset \mathcal{P}(\Omega)$. If $\mathcal{C}$ is a $\lambda$-system and $\mathcal{C}$ is a $\pi$-system, then $\mathcal{C}$ is a $\sigma$-algebra.

*Proof.* Suppose that $\mathcal{C}$ is a $\lambda$-system and $\mathcal{C}$ is a $\pi$-system. Then we need only verify the third axiom in the definition of a $\sigma$-algebra. Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{C}$. Define $B_1 = A_1$ and for $n \geq 2$, define $B_n = A_n \cap \left(\bigcup\limits_{k=1}^{n-1} A_k\right)^c = A_n \cap \left(\bigcap\limits_{k=1}^{n-1} A_k^c\right) \in \mathcal{C}$. Then $(B_n)_{n\in\mathbb{N}}$ is disjoint and therefore $\bigcup\limits_{n\in\mathbb{N}} A_n = \bigcup\limits_{n\in\mathbb{N}} B_n \in \mathcal{C}$. □

**Theorem 1.1.0.6.** (Dynkin's Theorem)
Let $\Omega$ be a set.

1. Let $\mathcal{P}$ be a $\pi$-system on $\Omega$ and $\mathcal{L}$ a $\lambda$-system on $\Omega$. If $\mathcal{P} \subset \mathcal{L}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.
2. Let $\mathcal{P}$ be a $\pi$-system on $\Omega$. Then $\sigma(\mathcal{P}) = \lambda(\mathcal{P})$

**Exercise 1.1.0.7.** Let $(\Omega, \mathcal{F})$ be a measurable space and $\mu, \nu$ probability measures on $(\Omega, \mathcal{F})$. Put $\mathcal{L}_{\mu,\nu} = \{A \in \mathcal{F} : \mu(A) = \nu(A)\}$. Then $\mathcal{L}_{\mu,\nu}$ is a $\lambda$-system on $\Omega$.

*Proof.*

1. $\varnothing \in \mathcal{L}_{\mu,\nu}$.
2. Let $A \in \mathcal{L}_{\mu,\nu}$. Then $\mu(A) = \nu(A)$. Thus

$$\begin{aligned}\mu(A^c) &= 1 - \mu(A) \\ &= 1 - \nu(A) \\ &= \nu(A^c)\end{aligned}$$

   So $A^c \in \mathcal{L}_{\mu,\nu}$.

3. Let $(A_n)_{n\in\mathbb{N}} \subset \mathcal{L}_{\mu,\nu}$. So for each $n \in \mathbb{N}$, $\mu(A_n) = \nu(A_n)$. Suppose that $(A_n)_{n\in\mathbb{N}}$ is disjoint. Then

$$\begin{aligned}\mu\bigg(\bigcup_{n\in\mathbb{N}} A_n\bigg) &= \sum_{n\in\mathbb{N}} \mu(A_n) \\ &= \sum_{n\in\mathbb{N}} \nu(A_n) \\ &= \nu\bigg(\bigcup_{n\in\mathbb{N}} A_n\bigg)\end{aligned}$$

   Hence $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{L}_{\mu,\nu}$.

□

**Exercise 1.1.0.8.** Let $(\Omega, \mathcal{F})$ be a measurable space, $\mu, \nu$ probability measures on $(\Omega, \mathcal{F})$ and $\mathcal{P} \subset \mathcal{F}$ a $\pi$-system on $\Omega$. Suppose that for each $A \in \mathcal{P}$, $\mu(A) = \nu(A)$. Then for each $A \in \sigma(\mathcal{P})$, $\mu(A) = \nu(A)$.

*Proof.* Using the previous exercise, we see that $\mathcal{P} \subset \mathcal{L}_{\mu,\nu}$. Dynkin's theorem implies that $\sigma(\mathcal{P}) \subset \mathcal{L}_{\mu,\nu}$. So for each $A \in \sigma(\mathcal{P})$, $\mu(A) = \nu(A)$. □

**Definition 1.1.0.9.** Let $F : \mathbb{R} \to \mathbb{R}$. Then $F$ is said to be a **probability distribution function** if

1. $F$ is right continuous
2. $F$ is increasing
3. $F(-\infty) = 0$ and $F(\infty) = 1$

**Definition 1.1.0.10.** Let $P$ be a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We define $F_P : \mathbb{R} \to \mathbb{R}$, by

$$F_P(x) = P((-\infty, x])$$

We call $F_P$ the **probability distribution function of** $P$.

**Exercise 1.1.0.11.** Let $(\Omega, \mathcal{F}, P)$ be a probability measure. Then $F_P$ is a probability distribution function.

*Proof.*

1. Let $x \in \mathbb{R}$ and $(x_n)_{n\in\mathbb{N}} \subset [x, \infty)$. Suppose that $x_n \to x$. Then $(x, x_n] \to \varnothing$ because $\limsup\limits_{n\to\infty}(x, x_n] = \varnothing$. Thus

$$F(x_n) - F(x) = P((x, x_n]) \to P(\varnothing) = 0$$

   This implies that

$$F(x_n) \to F(x)$$

   So $F$ is right continuous.

2. Clearly $F_P$ is increasing.

3. Continuity from below tells us that

$$F(-\infty) = \lim_{n\to-\infty} F(n) = \lim_{n\to-\infty} P((-\infty, n]) = 0$$

and continuity from above tell us that

$$F(\infty) = \lim_{n\to\infty} F(n) = \lim_{n\to\infty} P((-\infty, n]) = 1$$

□

**Exercise 1.1.0.12.** Let $\mu, \nu$ be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then $F_\mu = F_\nu$ iff $\mu = \nu$.

*Proof.* Clearly if $\mu = \nu$, then $F_\mu = F_\nu$. Conversely, suppose that $F_\mu = F_\nu$. Then for each $x \in \mathbb{R}$,

$$\begin{aligned}\mu((-\infty, x]) &= F_\mu(x) \\ &= F_\nu(x) \\ &= \nu((-\infty, x])\end{aligned}$$

Put $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$. Then $\mathcal{C}$ is a $\pi$-system and for each $A \in \mathcal{C}$, $\mu(A) = \nu(A)$. Hence for each $A \in \sigma(C) = \mathcal{B}(\mathbb{R})$, $\mu(A) = \nu(A)$. So $\mu = \nu$. □

**Definition 1.1.0.13.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X : \Omega \to \mathbb{R}^n$. Then $X$ is said to be a **random vector** on $(\Omega, \mathcal{F})$ if $X$ is $\mathcal{F}$-$\mathcal{B}(\mathbb{R}^n)$ measurable. If $n = 1$, then $X$ is said to be a **random variable**. We define

$$L^0_n(\Omega, \mathcal{F}, P) = \{X : \Omega \to \mathbb{R}^n : X \text{ is a random vector}\}$$

and

$$L^p_n(\Omega, \mathcal{F}, P) = \left\{X \in L^0_n : \int \|X\|^p dP < \infty\right\}$$

**Definition 1.1.0.14.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X$ a random variable on $(\Omega, \mathcal{F})$. We define the **probability distribution** of $X$, $P_X : \mathcal{B}(R) \to [0, 1]$, to be the measure

$$P_X = X_* P$$

That is, for each $A \in \mathcal{B}(\mathbb{R})$,

$$P_X(A) = P(X^{-1}(F))$$

We define the **probability distribution function** of $X$, $F_X : \mathbb{R} \to [0, 1]$, to be

$$F_X = F_{P_X}$$

**Definition 1.1.0.15.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X$ a random variable on $(\Omega, \mathcal{F})$. If $P_X \ll m$, we define the **probability density** of $X$, $f_X : \mathbb{R} \to \mathbb{R}$, by

$$f_X = \frac{dP_X}{dm}$$

**Exercise 1.1.0.16.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables on $(\Omega, \mathcal{F})$. Then for each $x \in \mathbb{R}$,

$$\P\left(\liminf_{n\to\infty} X_n > x\right) \leq \liminf_{n\to\infty} P(X_n > x)$$

*Proof.* Let $\omega \in \left\{\liminf\limits_{n\to\infty} X_n > x\right\}$. Then $x < \liminf\limits_{n\to\infty} X_n(\omega) = \sup\limits_{n\in\mathbb{N}}\left(\inf\limits_{k\geq n} X_k(\omega)\right)$. So there exists $n^* \in \mathbb{N}$ such that $x < \inf\limits_{k\geq n^*} X_k(\omega)$. Then for each $k \in \mathbb{N}$, $k \geq n^*$ implies that $x < X_k(\omega)$. So there exists $n^* \in \mathbb{N}$ such that for each $k \in \mathbb{N}$, $k \geq n^*$ implies that $\mathbf{1}_{\{X_k > x\}}(\omega) = 1$. Hence $\inf\limits_{k\geq n^*} \mathbf{1}_{\{X_k>x\}}(\omega) = 1$. Thus $\liminf\limits_{n\to\infty} \mathbf{1}_{\{X_k>x\}}(\omega) = \sup\limits_{n\in\mathbb{N}}\left(\inf_{k\geq n} \mathbf{1}_{\{X_k>x\}}(\omega)\right) = 1$. Therefore $\omega \in \liminf\limits_{n\to\infty}\{X_k > x\}$ and we have shown that

$$\left\{\liminf_{n\to\infty} X_n > x\right\} \subset \liminf_{n\to\infty}\{X_k > x\}$$

Then

$$P\left(\liminf_{n\to\infty} X_n > x\right) \leq P\left(\liminf_{n\to\infty}\{X_k > x\}\right)$$
$$\leq \liminf_{n\to\infty} P(\{X_k > x\})$$

□

**Definition 1.1.0.17.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^+(\Omega) \cup L^1$. Define the **expectation of X**, $E(X)$, to be

$$E(X) = \int X\, dP$$

.

## 1.2 Independence

**Definition 1.2.0.1.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{C} \subset \mathcal{F}$. Then $\mathcal{C}$ is said to be **independent** if for each $(A_i)_{i=1}^n \subset \mathcal{C}$,

$$P\left(\bigcap_{k=1}^n A_k\right) = \prod_{k=1}^n P(A_k)$$

**Definition 1.2.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{C}_1, \cdots, \mathcal{C}_n \subset \mathcal{F}$. Then $\mathcal{C}_1, \cdots, \mathcal{C}_n$ are said to be **independent** if for each $A_1 \in \mathcal{C}_1, \cdots, A_n \in \mathcal{C}_n$, $A_1, \cdots, A_n$ are independent.

**Note 1.2.0.3.** We will explicitely say that for each $i = 1, \cdots, n$, $\mathcal{C}_i$ is independent when talking about the independence of the elements of $\mathcal{C}_i$ to avoid ambiguity.

**Definition 1.2.0.4.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X_1, \cdots, X_2$ random variables on $(\Omega, \mathcal{F})$. Then $X_1, \cdots, X_n$ are said to be **independent** if for each $B_1, \cdots, B_n \in \mathcal{B}(\mathbb{R})$, $X_1^{-1}B_1, \cdots, X_n^{-1}B_n$ are independent.

**Exercise 1.2.0.5.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$. Then $X_1, \cdots, X_n$ are independent iff $\sigma(X_1), \cdots, \sigma(X_n)$ are independent.

*Proof.* Suppose that $X_1, \cdots, X_n$ are independent. Let $A_1, \in \sigma(X_1), \cdots, A_n \in \sigma(A_n)$. Then for each $i = 1, \cdots, n$, there exists $B_i \in \mathcal{B}(\mathbb{R})$ such that $A_i = X_i^{-1}(B_i)$. Then $A_1, \cdots, A_n$ are independent. Hence $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Conversely, suppose that $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Let $B_1, \cdots, B_n \in \mathcal{B}(\mathbb{R})$. Then for each $i = 1, \cdots, n$, $X_i^{-1}B_i \in \sigma(X_i)$. Then $X_1^{-1}B_1, \cdots, X_n^{-1}B_n$ are independent. Hence $X_1, \cdots, X_n$ are independent. □

**Exercise 1.2.0.6.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$ and $\mathcal{F}_1, \cdots, \mathcal{F}_n \subset \mathcal{F}$ a collection of $\sigma$-algebras on $\Omega$. Suppose that for each $i = 1, \cdots, n$, $X_i$ is $\mathcal{F}_i$-measurable. If $\mathcal{F}_1, \cdots, \mathcal{F}_n$ are independent, then $X_1, \cdots, X_n$ are independent.

*Proof.* For each $i = 1, \cdots, n$, $\sigma(X_i) \subset \mathcal{F}_i$. So $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Hence $X_1, \cdots, X_n$ are independent. □

**Exercise 1.2.0.7.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{C}_1, \cdots, \mathcal{C}_n \subset \mathcal{F}$. Suppose that for each $i = 1, \cdots, n$, $\mathcal{C}_i$ is a $\pi$-system and $\mathcal{C}_1, \cdots, \mathcal{C}_n$ are independent, then $\sigma(\mathcal{C}_1), \cdots, \sigma(\mathcal{C}_n)$ are independent.

*Proof.* Let $A_2 \in \mathcal{C}_2$. Define $\mathcal{L} = \{A \in \mathcal{F} : P(A \cap A_2) = P(A)P(A_2)\}$. Then

1. $\Omega \in \mathcal{L}$
2. If $A \in \mathcal{L}$, then

$$\begin{aligned} P(A^c \cap A_2) &= P(A_2) - P(A_2 \cap A) \\ &= P(A_2) - P(A_2)P(A) \\ &= (1 - P(A))P(A_2) \\ &= P(A^c)P(A_2) \end{aligned}$$

   So $A^c \in \mathcal{L}$

3. If $(B_n)_{n\in\mathbb{N}} \subset \mathcal{L}$ is disjoint, then

$$\begin{aligned}
P\bigg(\bigg[\bigcup_{n\in\mathbb{N}} B_n\bigg] \cap A_2\bigg) &= P\bigg(\bigcup_{n\in\mathbb{N}} B_n \cap A_2\bigg) \\
&= \sum_{n\in\mathbb{N}} P(B_n \cap A_2) \\
&= \sum_{n\in\mathbb{N}} P(B_n)P(A_2) \\
&= \bigg[\sum_{n\in\mathbb{N}} P(B_n)\bigg] P(A_2) \\
&= P\bigg(\bigcup_{n\in\mathbb{N}} A_n\bigg) P(A_2)
\end{aligned}$$

   So $\bigcup_{n\in\mathbb{N}} B_n \in \mathcal{L}$.

Thus $\mathcal{L}$ is a $\lambda$-system. Since $\mathcal{C}_1 \subset \mathcal{L}$ is a $\pi$-system, Dynkin's theorem tells us that $\sigma(\mathcal{C}_1) \subset \mathcal{L}$. Since $A_2 \in \mathcal{C}_2$ is arbitrary $\sigma(\mathcal{C}_1)$ and $\mathcal{C}_2$ are independent. The same reasoning implies that $\sigma(\mathcal{C}_1)$ and $\sigma(\mathcal{C}_2)$ are independent. Let $A_2 \in \mathcal{C}_1, \cdots, A_n \in \mathcal{C}_n$ We may do the same process with

$$\mathcal{L} = \bigg\{A \in \mathcal{F} : P\bigg(A \cap \bigg(\bigcap_{i=2}^{n} A_i\bigg)\bigg) = P(A)\prod_{i=2}^{n} P(A_i)\bigg\}$$

and conclude that $\sigma(\mathcal{C}_1), \mathcal{C}_2, \cdots, \mathcal{C}_n$ are independent. Which, using the same reasoning would imply that $\sigma(\mathcal{C}_1), \cdots, \sigma(\mathcal{C}_n)$ are independent. □

**Exercise 1.2.0.8.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$. Then $X_1, \cdots, X_n$ are independent iff for each $x_1, \cdots, x_n \in \mathbb{R}$,

$$P(X_1 \leq x_1, \cdots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i)$$

*Proof.* Suppose that $X_1, \cdots, X_n$ are independent. Then $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Let $x_1, \cdots, x_n \in \mathbb{R}$. Then for each $i = 1, \cdots, n$, $\{X_i \leq x_i\} \in \sigma(X_i)$. Hence
$P(X_1 \leq x_1, \cdots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i)$. Conversely, suppose that for each
$x_1, \cdots, x_n \in \mathbb{R}$, $P(X_1 \leq x_1, \cdots, X_n \leq x_n) = \prod_{i=1}^{n} P(X_i \leq x_i)$. Define $\mathcal{C} = \{(-\infty, x] : x \in \mathbb{R}\}$. Then $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{C})$. For each $i = 1, \cdots, n$, define $\mathcal{C}_i = X_i^{-1}\mathcal{C}$. Then for each $i = 1, \cdots, n$, $\mathcal{C}_i$ is a $\pi$-system and

$$\begin{aligned}
\sigma(\mathcal{C}_i) &= \sigma(X^{-1}(\mathcal{C})) \\
&= X_i^{-1}(\sigma(\mathcal{C})) \\
&= X_i^{-1}(\mathcal{B}(\mathbb{R})) \\
&= \sigma(X_i)
\end{aligned}$$

By assumption, $\mathcal{C}_1, \cdots, \mathcal{C}_n$ are independent. The previous exercies tells us that $\sigma(X_1), \cdots, \sigma(X_n)$ are independent. Then $X_1, \cdots, X_n$ are independent. □

**Exercise 1.2.0.9.** Let Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$. Define $X = (X_1, \cdots, X_n)$. If $X_1, \cdots, X_n$ are independent, then

$$P_X = \prod_{i=1}^{n} P_{X_i}$$

.

*Proof.* Let $A_1, \cdots, A_n \in \mathcal{B}(\mathbb{R})$. Then

$$\begin{aligned} P_X(A_1 \times \cdots \times A_n) &= P(X \in A_1 \times \cdots \times \in A_n) \\ &= P(X_1 \in A_1, \cdots, X_n \in A_n) \\ &= P(X_1 \in A_1) \cdots P(X_n \in A_n) \\ &= P_{X_1}(A_1) \cdots P_{X_n}(A_n) \\ &= \prod_{i=1}^n P_{X_i}(A_1 \times \cdots \times A_n) \end{aligned}$$

Put

$$\mathcal{P} = \{A_1 \times \cdots \times A_n : A_1 \in \mathcal{B}(R), \cdots, A_n \in \mathcal{B}(R)\}$$

Then $\mathcal{P}$ is a $\pi$-system and

$$\sigma(\mathcal{P}) = \mathcal{B}(R) \otimes \cdots \otimes \mathcal{B}(R) = \mathcal{B}(\mathbb{R}^n)$$

A previous exercise then tells us that $P_X = \prod\limits_{i=1}^n P_{X_i}$ □

**Exercise 1.2.0.10.** Let Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X_1, \cdots, X_n$ random variables on $(\Omega, \mathcal{F})$ and $f_1, \cdots, f_n : \mathbb{R} \to \mathbb{R} \in L^0$. Suppose that $f_1 \circ X_1, \cdots, f_n \circ X_n \in L^+(\Omega)$ or $f_1 \circ X_1, \cdots, f_n \circ X_n \in L^1(\Omega)$. If $X_1, \cdots, X_n$ are independent, then

$$E(f_1(X_1) \cdots f_n(X_n)) = \prod_{i=1}^n E(f_i(X_i))$$

*Proof.* Define the random vector $X : \Omega \to \mathbb{R}^n$ by $X = (X_1, \cdots, X_n)$ and $g : \mathbb{R}^n \to \mathbb{R}$ by $g(x_1, \cdots, x_n) = f_1(x_1) \cdots f_n(x_n)$. Suppose that for each $i = 1, \cdots, n$, $f_i \in L^+(\mathbb{R})$. Then $g \in L^+(\mathbb{R}^n)$ and by change of variables,

$$\begin{aligned} E(f_1(X_1) \cdots f_n(X_n)) &= E(g(X)) \\ &= \int_\Omega g \circ X \, dP \\ &= \int_{\mathbb{R}^n} g(x) \, dP_X(x) \\ &= \int_{R^n} g(x) d \prod_{i=1}^n P_{X_i}(x) \\ &= \prod_{i=1}^n \int_{\mathbb{R}} f_i(x) dP_{X_i}(x) \\ &= \prod_{i=1}^n \int_\Omega f_i \circ X dP \\ &= \prod_{i=1}^n E(f_i(X_i)) \end{aligned}$$

If for each $i = 1, \cdots, n$, $f_i \in L^1(\mathbb{R}, P_{X_i})$, then following the above reasoning with $|g|$ tells us that $g \in L^1(\mathbb{R}^n, P_X)$ and we use change of variables and Fubini's theorem to get the same result. □

## 1.3 $L^p$ Spaces for Probability

**Note 1.3.0.1.** Recall that for a probability space $(\Omega, \mathcal{F}, P)$ and $1 \leq p \leq q \leq \infty$ we have $L^q \subset L^p$ and for each $X \in L^q$, $\|X\|_p \leq \|X\|_q$. Also recall that for $X, Y \in \mathrm{L}^2$, we have that $\|XY\|_1 \leq \|X\|_2 \|X\|_2$.

**Definition 1.3.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^2$. Define the **variance of X**, $Var(X)$, to be

$$Var(X) = E(\big[\big(X - E(X)\big)^2\big)$$

.

**Definition 1.3.0.3.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. Define the

**Definition 1.3.0.4.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. Define the **covariance of** $X$ **and** $Y$, $Cov(X,Y)$, to be

$$Cov(X,Y) = E([X - E(X)][Y - E(Y)])$$

**Exercise 1.3.0.5.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. Then the covariance is well defined and $Cov(X,Y)^2 \leq Var(X)Var(Y)$

*Proof.* By Holder's inequality,

$$\begin{aligned}
|Cov(X,Y)| &= \left| \int (X - E(X))(Y - E(Y))\, dP \right| \\
&\leq \int |(X - E(X))(Y - E(Y))|\, dP \\
&= \|(X - E(X))(Y - E(Y))\|_1 \\
&\leq \|X - E(X)\|_2 \|(Y - E(Y)\|_2 \\
&= \left( \int |X - E(X)|^2\, dP \right)^{\frac{1}{2}} \left( |Y - E(Y)|^2 \right)^{\frac{1}{2}} \\
&= Var(X)^{\frac{1}{2}} Var(Y)^{\frac{1}{2}}
\end{aligned}$$

So $Cov(X,Y)^2 \leq Var(X)Var(Y)$. □

**Exercise 1.3.0.6.** Let $(\Omega, \mathcal{F}, P)$ be a measure space and $X, Y \in L^2$. Then

1. $Cov(X,Y) = E(XY) - E(X)E(Y)$
2. If $X, Y$ are independent, then $Cov(X,Y) = 0$
3. $Var(X) = E(X^2) - E(X)^2$
4. for each $a, b \in \mathbb{R}$, $Var(aX + b) = a^2 Var(X)$.
5. $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X,Y)$

*Proof.*

1. We have that

$$\begin{aligned}
Cov(X,Y) &= E\Big[(X - E(X))(Y - E(Y))\Big] \\
&= E(XY - E(Y)X - E(X)Y + E(X)E(Y)) \\
&= E(XY) - E(X)E(Y) - E(X)E(Y) + E(X)E(Y) \\
&= E(XY) - E(X)E(Y)
\end{aligned}$$

2. Suppose that $X, Y$ are independent. Then $E(XY) = E(X)E(Y)$. Hence

$$\begin{aligned}
Cov(X,Y) &= E(XY) - E(X)E(Y) \\
&= E(X)E(Y) - E(X)E(Y) \\
&= 0
\end{aligned}$$

3. Part (1) implies that

$$\begin{aligned}
Var(X) &= Cov(X,X) \\
&= E(X^2) - E(X)^2
\end{aligned}$$

4. Let $a, b \in \mathbb{R}$. Then

$$\begin{aligned} Var(aX+b) &= E[(aX+b)^2] - E(aX+b)^2 \\ &= E[a^2X^2 + 2abX + b^2] - (aE(X)+b)^2 \\ &= a^2E(X^2) + 2abE(X) + b^2 - (a^2E(X)^2 + 2abE(X) + b^2) \\ &= a^2(E(X^2) - E(X)^2) \\ &= a^2 Var(X) \end{aligned}$$

5. We have that

$$\begin{aligned} Var(X+Y) &= E[(X+Y)^2] - E[X+Y]^2 \\ &= E[X^2 + 2XY + Y^2] - (E(X) + E[Y])^2 \\ &= E(X^2) + 2E[XY] + E[Y^2] - (E(X)^2 + 2E(X)E[Y] + E[Y]^2) \\ &= (E(X^2) - E(X)^2) + (E[Y^2] - E[Y]^2) + 2(E[XY] - E(X)E[Y]) \\ &= Var(X) + Var(Y) + 2Cov(X,Y) \end{aligned}$$

□

**Definition 1.3.0.7.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^2$. The **correlation of X and Y**, $Cor(X,Y)$, is defined to be

$$Cor(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

**Exercise 1.3.0.8.**

**Exercise 1.3.0.9. Jensen's Inequality:**
Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X \in L^1$ and $\phi : \mathbb{R} \to \mathbb{R}$. If $\phi$ is convex, then

$$\phi(E(X)) \leq E[\phi(X)]$$

*Proof.* Put $x_0 = E(X)$. Since $\phi$ is convex, there exist $a, b \in \mathbb{R}$ such that $\phi(x_0) = ax_0 + b$ and for each $x \in \mathbb{R}$, $\phi(x) \geq ax + b$. Then

$$\begin{aligned} E[\phi(X)] &= \int \phi(X)\, dP \\ &\geq \int [aX + b]\, dP \\ &= a\int X\, dP + b \\ &= aE(X) + b \\ &= ax_0 + b \\ &= \phi(x_0) \\ &= \phi(E(X)) \end{aligned}$$

□

**Exercise 1.3.0.10.** Markov's Inequality: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^+$. Then for each $a \in (0, \infty)$,

$$P(X \geq a) \leq \frac{E(X)}{a}$$

*Proof.* Let $a \in (0, \infty)$. Then $a\mathbf{1}_{\{X \geq a\}} \leq X\mathbf{1}_{\{X \geq a\}}$. Thus

$$\begin{aligned} aP(X \geq a) &= \int a\mathbf{1}_{\{X \geq a\}}\, dP \\ &= \int X\mathbf{1}_{\{X \geq a\}}\, dP \\ &\leq \int X\, dP \\ &= E(X) \end{aligned}$$

Therefore

$$P(X \geq a) \leq \frac{E(X)}{a}$$

. □

**Exercise 1.3.0.11.** Chebychev's Inequality: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^2$. Then for each $a \in (0, \infty)$,

$$P(|X - E(X)| \geq a) \leq \frac{Var(X)}{a^2}$$

*Proof.* Let $a \in (0, \infty)$. Then

$$\begin{aligned} P(|X - E(X)| \geq a) &= P((X - E(X))^2 \geq a^2) \\ &\leq \frac{E[(X - E(X))^2]}{a^2} \\ &= \frac{Var(X)}{a^2} \end{aligned}$$

□

**Exercise 1.3.0.12.** Chernoff's Bound: Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X \in L^2$. Then for each $a, t \in (0, \infty)$,

$$P(X \geq a) \leq e^{-ta}E[e^{tX}]$$

*Proof.* Let $a, t \in (0, \infty)$. Then

$$\begin{aligned} P(X \geq a) &= P(tX \geq ta) \\ &= P(e^{tX} \geq e^{ta}) \\ &\leq e^{-ta}E[e^{tX}] \end{aligned}$$

□

**Exercise 1.3.0.13.** Weak Law of Large Numbers: Let $(\Omega, \mathcal{F}, P)$ be a probability space $(X_i)_{i \in \mathbb{N}} \subset L^2$. Suppose that $(X_i)_{i \in \mathbb{N}}$ are iid. Then

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} E[X_1]$$

*Proof.* Put $\mu = E[X_1]$ and $\sigma^2 = Var(X_1)$. Then

$$\begin{aligned} E[\frac{1}{n}\sum_{i=1}^{n} X_i] &= \frac{1}{n}\sum_{i=1}^{n} E[X_i] \\ &= \frac{1}{n}\sum_{i=1}^{n} \mu \\ &= \mu \end{aligned}$$

and

$$
\begin{aligned}
Var(\frac{1}{n}\sum_{i=1}^{n} X_i) &= \frac{1}{n^2} Var(\sum_{i=1}^{n} X_i) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) \\
&= \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 \\
&= \frac{\sigma^2}{n}
\end{aligned}
$$

Let $\epsilon > 0$. Then

$$
\begin{aligned}
P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - E[X_1]\right| \geq \epsilon\right) &= P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu\right| \geq \epsilon\right) \\
&= P\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - E(\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]\right| \geq \epsilon\right) \\
&\leq \frac{Var(\frac{1}{n}\sum_{i=1}^{n} X_i)}{\epsilon^2} \\
&= \frac{\sigma^2/n}{\epsilon^2} \\
&= \frac{\sigma^2}{n\epsilon^2} \to 0
\end{aligned}
$$

So

$$
\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{P} E[X_1]
$$

□

## 1.4 Borel Cantelli Lemma

**Exercise 1.4.0.1. Borel Cantelli Lemma:**
Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(A_n)_{n\in\mathbb{N}} \subset \mathcal{F}$.

1. If $\sum\limits_{n\in\mathbb{N}} P(A_n) < \infty$, then $P(\limsup\limits_{n\to\infty} A_n) = 0$.
2. If $(A_n)_{n\in\mathbb{N}}$ are independent and $\sum\limits_{n\in\mathbb{N}} P(A_n) = \infty$, then $P(\limsup\limits_{n\to\infty} A_n) = 1$

*Proof.*

1. Suppose that $\sum\limits_{n\in\mathbb{N}} P(A_n) < \infty$. Recall that
$$\limsup_{n\to\infty} A_n = \left\{\omega \in \Omega : \sum_{n\in\mathbb{N}} 1_{A_n}(\omega) = \infty\right\}$$
Then
$$\begin{aligned}\infty &> \sum_{n\in\mathbb{N}} P(A_n) \\ &= \sum_{n\in\mathbb{N}} \int 1_{A_n}\, dP \\ &= \int \sum_{n\in\mathbb{N}} 1_{A_n}\, dP\end{aligned}$$
Thus $\sum\limits_{n\in\mathbb{N}} 1_{A_n} < \infty$ a.e. and $P(\limsup\limits_{n\to\infty} A_n) = 0$.
2. Suppose that $(A_n)_{n\in\mathbb{N}}$ are independent and $\sum\limits_{n\in\mathbb{N}} P(A_n) = \infty$.

$\square$

**Exercise 1.4.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(X_n)_{n\in\mathbb{N}} \subset L^0$ and $X \in L^0$.

1. If for each $\epsilon > 0$, $\sum\limits_{n\in\mathbb{N}} P(|X_n - X| \geq \epsilon) < \infty$, then $X_n \to X$ a.e.
2. If $(X_n)_{n\in\mathbb{N}}$ are independent and there exists $\epsilon > 0$ such that $\sum\limits_{n\in\mathbb{N}} P(|X_n - X| \geq \epsilon) = \infty$, then $X_n \not\to X$ a.e.

*Proof.*

1. For $\epsilon > 0$ and $n \in \mathbb{N}$, set $A_n(\epsilon) = \{\omega \in \Omega : |X_n(\omega) - X(\omega)| \geq \epsilon\}$. Suppose that for each $\epsilon > 0$, $\sum\limits_{n\in\mathbb{N}} P(|X_n - X| \geq \epsilon) < \infty$. The Borel-Cantelli lemma implies that for each $m \in \mathbb{N}$,
$$P(\limsup_{n\to\infty} A_n(1/m)) = 0$$
Let $\omega \in \Omega$. Then $X_n(\omega) \not\to X(\omega)$ iff
$$\omega \in \bigcup_{m\in\mathbb{N}} \limsup_{n\to\infty} A_n(1/m)$$
So
$$\begin{aligned}P(X_n \not\to X) &= P\bigg(\bigcup_{m\in\mathbb{N}} \limsup_{n\to\infty} A_n(1/m)\bigg) \\ &\leq \sum_{m\in\mathbb{N}} P(\limsup_{n\to\infty} A_n(1/m)) \\ &= 0\end{aligned}$$
Hence $X_n \to X$ a.e.

2.

$\square$

# Chapter 2

# Probability on locally compact Groups

**Note 2.0.0.1.** In this section, familiarity with Haar measure will be assumed. This section is intended as a continuation of section 7 of [4].

## 2.1 Action on Probability Measures

**Note 2.1.0.1.** We recall some notation from section 7.1 of [4].

- $l_g \in \mathrm{Homeo}(G)$, $l_g(x) = gc$
- $L_g \in \mathrm{Sym}(L_0(G))$, $L_g f = f \circ l_g^{-1}$ We continue from section 7

**Note 2.1.0.2.** The next exercise generalizes the notion of a scale-family.

**Exercise 2.1.0.3.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $G$ a locally compact group, $\mu$ a left Haar measure on $G$, $X \in L^0_G$ and $g \in G$. If $P_X \ll \mu$, then $f_{gX} = L_g f_X$.

*Proof.* Suppose that $P_X \ll \mu$. Let $A \in \mathcal{B}(G)$. Then

$$\begin{aligned} P_{gX}(A) &= P(gX \in A) \\ &= P(X \in g^{-1}A) \\ &= P_X(g^{-1}A) \\ &= P_X(l_g^{-1}(A)) \\ &= l_{g*}P_X(A) \\ &= g \cdot P_X(A) \end{aligned}$$

The previous exercise tells us that $f_{gX} = L_g f_X$. □

# Chapter 3

# Gaussian Measures

## 3.1 Gaussian Measures on $\mathbb{R}$

**Definition 3.1.0.1.** Let $\mu \in \mathcal{M}_1(\mathbb{R})$. Then $\mu$ is said to be **Gaussian** if

## 3.2 Gaussian Measures on Topological Vector Spaces

**Definition 3.2.0.1.** Let $W$ be a topological vector space and $\mu \in \mathcal{M}_1(W)$. Then $\mu$ is said to be **Gaussian** if for each $\phi \in W^*$, $\phi_*\mu$ is Gaussian.

need exercise in analysis notes showing $\mathbb{R}^{\mathbb{N}}$ is a topological vector space.

**Definition 3.2.0.2.** For $n \in \mathbb{N}$, define $\mu_n \in \mathcal{M}_1(\mathbb{R})$ by

$$\mu_n(A) := \int_A \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dm(x).$$

An exercise in measure theory notes section on projective limits of measure spaces implies that there exists a unique $\mu \in \mathcal{M}_1(\mathbb{R}^{\mathbb{N}})$ such that

- for each $n \in \mathbb{N}$, $\pi_n^*\mu = \mu_n$ and for each,
- for each $(A_n)_{n\in\mathbb{N}} \in \mathcal{B}(\mathbb{R})^{\mathbb{N}}$,

$$\mu\bigg(\prod_{n\in\mathbb{N}} A_n\bigg) = \prod_{n\in\mathbb{N}} \mu_n(A_n).$$

We refer to $\mu$ as the **standard Gaussian measure** on $\mathbb{R}^{\mathbb{N}}$.

**Exercise 3.2.0.3.** $\mu$ is Gaussian.

*Proof.* Let $n \in \mathbb{N}$ and $\lambda \in \mathbb{R}$. Define $\eta_\lambda : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}$ by $\eta_\lambda(x) := \lambda x$. Then

$$\begin{aligned}(\lambda\pi_n)_*\mu &= (\eta_\lambda \circ \pi_n)_*\mu \\ &= (\eta_\lambda)_*((\pi_n)_*\mu) \\ &= (\eta_\lambda)_*\mu_n\end{aligned}$$

Define $\nu \in \mathcal{M}_1(\mathbb{R})$ by $\nu := (\eta_\lambda)_*\mu_n$. If $\lambda = 0$, then $\nu = \delta_0$. Suppose that $\lambda \neq 0$. Since $\mu_n \ll m$, (check/give details) we have that $\nu \ll m$. The Radon-Nikodym-Lebesgue theorem implies that there exists $f \in L^1(\mathbb{R}, \mathcal{B}(R), m)$ such that $d\nu = f\, dm$. An exercise in analysis notes (need to rework in the newer general framerwork) implies that for $m$-a.e. $x \in \mathbb{R}$,

$$\widehat{\nu}(x) = \int_{\mathbb{R}} e^{itx}\, d\nu(t) \qquad =$$

$$\begin{aligned}f(x) &= \lim_{r\to 0} \frac{\nu((x-r, x+r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{\mu_n(\eta_\lambda^{-1}(x-r, x+r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{\mu_n(\lambda^{-1}(x-r, x+r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{\mu_n((\lambda^{-1}x - \lambda^{-1}r, \lambda^{-1}x + \lambda^{-1}r))}{m((x-r, x+r))} \\ &= \lim_{r\to 0} \frac{1}{2r} \int_{\lambda^{-1}x-\lambda^{-1}r}^{\lambda^{-1}x+\lambda^{-1}r} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dm(x) \\ &= \lim_{r\to 0} \frac{1}{2r} \int_u\end{aligned}$$

$f(x)$ Let $\phi \in (\mathbb{R}^{\mathbb{N}})^*$. An exercise in the analysis notes section on topological vector spaces subsection on products implies that $(\mathbb{R}^{\mathbb{N}})^* = \operatorname{span}\{\pi_j : j \in \mathbb{N}\}$.

Then show that the pushforward of a linear combos of projection maps is Gaussian $\square$

## 3.3 Gaussian Measures on Hilbert Spaces

**Note 3.3.0.1.** recall definitions of $(Y, (\tau_n)_{n\in\mathbb{N}})$ and $((Y_n)_{n\in\mathbb{N}}, (\tau_{n,k})_{(n\leq k)})$ from measure and integration notes section on projective systems of complex measure spaces, specifically the exercise on existence and uniqueness of product measures. We still need an exercise to show that $(Y, (\tau_n)_{n\in\mathbb{N}})$ is a Top-projective limit of $((Y_n)_{n\in\mathbb{N}}, (\tau_{n,k})_{(n\leq k)})$ in the analysis notes.

**Exercise 3.3.0.2.** Let $a \in \mathbb{C}^{\mathbb{N}}$. Then $l^{2,a} \in \mathcal{B}(\mathbb{R}^{\mathbb{N}})$.

*Proof.* We $\square$

Chapter 4

# Weak Convergence of Measures

# Chapter 5

# Concentration Inequalities

## 5.1 Introduction

**Exercise 5.1.0.1.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^0_{\mathbb{R}}(\Omega, \mathcal{F}, P)$. Then for each $s, t \in \mathbb{R}$,

$$P(X + Y \geq s + t) \leq P(X \geq s) + P(Y \geq t)$$

*Proof.* For $Z \in L^0_{\mathbb{R}}(\Omega, \mathcal{F}, P)$ and $t \in \mathbb{R}$, define $A^t_Z \in \mathcal{F}$ by

$$A^t_Z = \{\omega \in \Omega : Z(\omega) \geq t\}$$

Let $s, t \in \mathbb{R}$. Since $(A^s_X)^c \cap (A^t_Y)^c \subset (A^{s+t}_{X+Y})^c$, we have that $A^{s+t}_{X+Y} \subset A^s_X \cup A^t_Y$. Then

$$\begin{aligned} P(X + Y \geq s + t) &= P(A^{s+t}_{X+Y}) \\ &\leq P(A^s_X) + P(A^t_Y) \\ &= P(X \geq s) + P(Y \geq t) \end{aligned}$$

□

**Exercise 5.1.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X, Y \in L^+(\Omega, \mathcal{F}, P)$. Then for each $s, t \geq 0$,

$$P(XY \geq st) \leq P(X \geq s) + P(Y \geq t)$$

*Proof.* For $Z \in L^0_{\mathbb{R}}(\Omega, \mathcal{F}, P)$ and $t \in \mathbb{R}$, define define $A^t_Z \in \mathcal{F}$ by

$$A^t_Z = \{\omega \in \Omega : Z(\omega) \geq t\}$$

Let $s, t \in \mathbb{R}$. Since $(A^s_X)^c \cap (A^t_Y)^c \subset (A^{st}_{XY})^c$, we have that $A^{st}_{XY} \subset A^s_X \cup A^t_Y$. Then

$$\begin{aligned} P(XY \geq st) &= P(A^{st}_{XY}) \\ &\leq P(A^s_X) + P(A^t_Y) \\ &= P(X \geq s) + P(Y \geq t) \end{aligned}$$

□

## 5.2 Sub $\alpha$-Exponential Random Variables

**Definition 5.2.0.1.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X \in L^0(\Omega, \mathcal{F}, P)$ and $\alpha > 0$. Then $X$ is said to be **sub $\alpha$-exponential** if there exist $M, K > 0$ such that for each $t \geq 0$,

$$P(|X| \geq t) \leq Me^{-Kt^\alpha}$$

**Exercise 5.2.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X \in L^0(\Omega, \mathcal{F}, P)$ and $\alpha > 0$. Then the following are equivalent:

1. $X$ is sub $\alpha$-exponential
2. there exists $K > 0$ such that for each $p \geq 1$, $\|X\|_p \leq Kp^{1/\alpha}$
3. 
4. 

*Proof.*

- (1) $\implies$ (2):
  Choose $C_\alpha > 0$ such that for each $x \geq \alpha^{-1}$, $\Gamma(x) \leq C_\alpha x^x$. Since $X$ is sub $\alpha$-exponential, there exist $M, K_0 > 0$ such that for each $t \geq 0$, $P(|X| \geq t) \leq Me^{-Kt^\alpha}$. Set $K = \max(M\alpha^{-1}C_\alpha, 1)2K_0^{-1/\alpha}\alpha^{-1/\alpha}$. Let $p \geq 1$. Then $p\alpha^{-1} \geq \alpha^{-1}$

$$\begin{aligned}
\|X\|_p^p &= E(|X|^p) \\
&= \int_0^\infty P(|X|^p \geq t)dt \\
&= \int_0^\infty P(|X| \geq t^{1/p})dt \\
&\leq \int_0^\infty Me^{-K_0 t^{\alpha/p}} dt \\
&= Mp\alpha^{-1} \int_0^\infty u^{p/\alpha - 1} e^{-K_0 u} du \\
&= Mp\alpha^{-1}\Gamma(p/\alpha)K_0^{-p/\alpha} \\
&\leq Mp\alpha^{-1}C_\alpha(p\alpha^{-1})^{p/\alpha}K_0^{-p/\alpha}
\end{aligned}$$

  Therefore

$$\begin{aligned}
\|X\|_p &\leq (M\alpha^{-1}C_\alpha)^{1/p}p^{1/p}K_0^{-1/\alpha}\alpha^{-1/\alpha}p^{1/\alpha} \\
&\leq \max(M\alpha^{-1}C_\alpha, 1)2K_0^{-1/\alpha}\alpha^{-1/\alpha}p^{1/\alpha} \\
&= Kp^{1/\alpha}
\end{aligned}$$

- (2) $\implies$ (3):

$\square$

**Definition 5.2.0.3.** Let $\psi : [0, \infty) \to [0, \infty)$. Then $\psi$ is said to be an **Orlicz function** if

1. $\psi$ is convex
2. $\psi$ is increasing
3. $\psi(x) \to \infty$ as $x \to \infty$
4. $\psi(0) = 0$

**Definition 5.2.0.4.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\psi : [0, \infty) \to [0, \infty)$ and Orlicz function. We define the **Orlicz $\psi$-norm**, denoted $\|\cdot\|_\psi : L^0(\Omega, \mathcal{F}, P) \to [0, \infty]$, by

$$\|X\|_\psi = \inf\{t > 0 : E[\psi(|X|/t)] \leq 1\}$$

We define the **Orlicz $\psi$-space**, denoted $L^\psi(\Omega, \mathcal{F}, P)$, by

$$L^\psi(\Omega, \mathcal{F}, P) = \{X \in L^1(\Omega, \mathcal{F}, P) : \|X\|_\psi < \infty\}$$

**Exercise 5.2.0.5.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\psi : [0,\infty) \to [0,\infty)$ an Orlicz function. Then $L^\psi$ is a vector space and $\|\cdot\|_\psi : L^\psi \to [0,\infty)$ is a norm.
**Hint:** note that

- for $s, t > 0$,
$$\frac{|X|}{s+t} + \frac{|Y|}{s+t} = \frac{s}{s+t}\frac{|X|}{s} + \frac{t}{s+t}\frac{|Y|}{t}$$
- $\psi$ is star-shaped, i.e. for each $x, t \geq 0$ and $f(tx) \leq tf(x)$.

*Proof.* For $X \in L^0(\Omega, \mathcal{F}, P)$, define $A_X \in \mathcal{F}$ by

$$A_X = \{t > 0 : E[\psi(|X|/t)] \leq 1\}$$

Let $X, Y \in L^\psi(\Omega, \mathcal{F}, P)$ and $\lambda \in \mathbb{C}$. Since $\|X\|_\psi < \infty$ and $\|Y\|_\psi < \infty$, we have that $A_X \neq \varnothing$ and $A_Y \neq \varnothing$,

1. **subadditivity:** Let $\epsilon > 0$. Then there exists $s \in A_X$ and $t \in A_Y$ such that $s < \inf A_X + \epsilon/2$ and $t < \inf A_Y + \epsilon/2$. Since is convex and increasing, we have that
$$\begin{aligned}\psi\bigg(\frac{|X+Y|}{s+t}\bigg) &\leq \psi\bigg(\frac{|X|}{s+t} + \frac{|Y|}{s+t}\bigg) \\ &= \psi\bigg(\frac{s}{s+t}\frac{|X|}{s} + \frac{t}{s+t}\frac{|Y|}{t}\bigg) \\ &\leq \frac{s}{s+t}\psi\bigg(\frac{|X|}{s}\bigg) + \frac{t}{s+t}\psi\bigg(\frac{|Y|}{t}\bigg)\end{aligned}$$

   Therefore,
$$\begin{aligned}E\bigg[\psi\bigg(\frac{|X+Y|}{s+t}\bigg)\bigg] &\leq \frac{s}{s+t}E\bigg[\psi\bigg(\frac{|X|}{s}\bigg)\bigg] + \frac{t}{s+t}E\bigg[\psi\bigg(\frac{|Y|}{t}\bigg)\bigg] \\ &\leq \frac{s}{s+t} + \frac{t}{s+t} \\ &\leq 1\end{aligned}$$

   Hence $s + t \in A_{X+Y}$. Thus $A_{X+Y} \neq \varnothing$. Since $s < \inf A_X + \epsilon/2$ and $t < \inf A_Y + \epsilon/2$, we have that
$$\begin{aligned}\|X+Y\|_\psi &= \inf A_{X+Y} \\ &\leq s+t \\ &< \inf A_X + \inf A_Y + \epsilon \\ &= \|X\|_\psi + \|Y\|_\psi + \epsilon\end{aligned}$$

   Since $\epsilon > 0$ is arbitrary, $\|X+Y\|_\psi \leq \|X\|_\psi + \|Y\|_\psi$. So $X + Y \in L^\psi(\Omega, \mathcal{F}, P)$

2. **absolute homogeneity:** Suppose that $\lambda = 0$. Then for each $t > 0$,
$$\begin{aligned}E[\psi(|\lambda X|/t)] &= E[\psi(0)] \\ &= E[0] \\ &= 0 \\ &\leq 1\end{aligned}$$

   Thus
$$\begin{aligned}\|\lambda X\|_\psi &= 0 \\ &= |\lambda|\|X\|_\psi\end{aligned}$$

Suppose that $\lambda \neq 0$. Since for each $t > 0$,

$$\psi\left(\frac{|X|}{t}\right) = \psi\left(\frac{|\lambda X|}{|\lambda| t}\right)$$

we have that for each $t > 0$, $t \in A_X$ iff $|\lambda| t \in A_{\lambda X}$. Therefore $A_{\lambda X} = |\lambda| A_X$ and

$$\begin{aligned}\|\lambda X\|_\psi &= \inf A_{\lambda X} \\ &= |\lambda| \inf A_X \\ &= |\lambda| \|X\|_\psi\end{aligned}$$

So $|\lambda X\| \in L^\psi(\Omega, \mathcal{F}, P)$.

3. **positive definiteness:** Note that since $\psi$ is increasing, for each $t_0 > 0$, $t_0 \in A_X$ implies that $[t_0, \infty) \subset A_X$. Suppose that $\|X\|_\psi = 0$. Then $(0, \infty) \subset A_X$. Jensen's inequality implies that for each $t > 0$,

$$\begin{aligned}\psi\left(\frac{E|X|}{t}\right) &\leq E\left[\psi\left(\frac{|X|}{t}\right)\right] \\ &\leq 1\end{aligned}$$

For the sake of contradiction, suppose that $E|X| > 0$. Since $\psi(x) \to \infty$ as $x \to \infty$, Choose $x > 0$ such that $\psi(x) > 1$. Set $t = E|X|/x$. Then $t > 0$ and

$$\begin{aligned}1 &< \psi(x) \\ &= \psi\left(\frac{E|X|}{t}\right) \\ &\leq 1\end{aligned}$$

which is a contradiction. Hence $E|X| = 0$. Thus $X = 0$ a.s.

$\square$

**Exercise 5.2.0.6.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\psi : [0, \infty) \to [0, \infty)$ an Orlicz function. Then $L^\psi(\Omega, \mathcal{F}, P)$ is a Banach space.

*Proof.* $\square$

# Chapter 6

# Conditional Expectation and Probability

## 6.1 Conditional Expectation

**Exercise 6.1.0.1.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ a sub $\sigma$-algebra of $\mathcal{F}$ and $X \in L^1(\Omega, \mathcal{F}, P)$. Define $P_{\mathcal{G}} = P|_{\mathcal{G}}$ and $Q : \mathcal{G} \to [0, \infty)$ by $Q(G) = \int_G X dP$. Then $Q \ll P_{\mathcal{G}}$.

*Proof.* Let $G \in \mathcal{G}$. Suppose that $P_{\mathcal{G}}(G) = 0$. By definition, $P(G) = 0$. So $Q(G) = 0$ and $Q \ll P_{\mathcal{G}}$. □

**Definition 6.1.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ a sub $\sigma$-algebra of $\mathcal{F}$ and $X, Y \in L^1(\Omega, \mathcal{F}, P)$. Then $Y$ is said to be a **conditional expectation of $X$ given $\mathcal{G}$** if

1. $Y$ is $\mathcal{G}$-measurable
2. for each $G \in \mathcal{G}$,

$$\int_G Y\, dP = \int_G X\, dP$$

Since (2) implies that conditional expectations of $X$ given $\mathcal{G}$ are equal $P_{\mathcal{G}}$-a.e., we write $Y = E(X|\mathcal{G})$.

**Note 6.1.0.3.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $(S, \mathcal{S})$ a measurable space, $X \in L^1(\Omega, \mathcal{F}, P)$ and $Y \in L^0_S(\Omega, \mathcal{F})$. We typically write $E(X|Y)$ instead of $E(X|Y^*\mathcal{S})$.

**Exercise 6.1.0.4. Existence of Conditional Expectation:**
Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{G}$ a sub $\sigma$-algebra of $\mathcal{F}$ and $X \in L^1(\Omega, \mathcal{F}, P)$. Define $Q$ and $P_{\mathcal{G}}$ as in the previous exercise. Define $Y = dQ/dP_{\mathcal{G}}$. Then $Y$ is a conditional expectation of $X$ given $\mathcal{G}$.

*Proof.* The Radon-Nikodym theorem implies that $Y$ is $\mathcal{G}$-measurable. Since $Q$ is finite, so is $|Q|$. Since $d|Q| = |Y|\, dP_{\mathcal{G}}$, we have that $Y \in L^1(\Omega, \mathcal{G}, P_{\mathcal{G}})$. An exercise in section 3.3 of [4], implies that for each $G \in \mathcal{G}$

$$\begin{aligned}\int_G Y\, dP &= \int_G Y\, dP_{\mathcal{G}} \\ &= Q(G) \\ &= \int_G X\, dP\end{aligned}$$

□

**Definition 6.1.0.5.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $(S, \mathcal{S})$ a measurable space, $X \in L^1(\Omega, \mathcal{F}, P)$ and $Y \in L^0_S(\Omega, \mathcal{F})$. Let $\phi \in L^0(Y(\Omega), \mathcal{S} \cap Y(\Omega))$. Then $\phi$ is said to be a **conditional expectation function of $X$ given $Y$** if for each $B \in \mathcal{S} \cap Y(\Omega)$,

$$\int_{Y^{-1}(B)} X\, dP = \int_B \phi\, dP_Y$$

To denote this, we write $\phi(y) = E[X|Y = y]$.

**Exercise 6.1.0.6. Existence of Conditional Expectation Function:**
Let $(\Omega, \mathcal{F}, P)$ be a probability space, $(S, \mathcal{S})$ a measurable space, $X \in L^1(\Omega, \mathcal{F}, P)$ and $Y \in L^0_S(\Omega, \mathcal{F})$. Suppose that for each $y \in S$, $\{y\} \in \mathcal{S}$. Then there exists $\phi \in L^0(Y(S), \mathcal{S} \cap Y(\Omega))$ such that $\phi$ is a conditional expectation function of $X$ given $Y$.
**Hint:** Doob-Dynkin lemma

*Proof.* Since $E[X|Y] \in L^0(\Omega, Y^*\mathcal{S})$, the Doob-Dynkin lemma implies that there exists $\phi \in L^0(Y(\Omega), \mathcal{S} \cap Y(\Omega))$ such that $\phi \circ Y = E(X|Y)$. Let $B \in \mathcal{S} \cap Y(\Omega)$. Then

$$\begin{aligned}\int_B \phi \, dP_Y &= \int_{Y^{-1}(B)} \phi \circ Y \, dP \\ &= \int_{Y^{-1}(B)} E(X|Y) \, dP \\ &= \int_{Y^{-1}(B)} X \, dP\end{aligned}$$

□

## 6.2 Conditional Probability

**Definition 6.2.0.1.** Let $(A, \mathcal{A})$ be a measurable space, $(B, \mathcal{B}, P_Y)$ a probability space and $Q : B \times \mathcal{A} \to [0, 1]$. Then $Q$ is said to be a **stochastic transition kernel from** $(B, \mathcal{B}, P)$ **to** $(A, \mathcal{A})$ if

1. for each $E \in \mathcal{A}$, $Q(\cdot, E)$ is $\mathcal{B}$-measurable

2. for $P$-a.e. $b \in B$, $Q(b, \cdot)$ is a probability measure on $(A, \mathcal{A})$

**Definition 6.2.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n(\Omega, \mathcal{F}, P)$ and $Q : \mathbb{R}^n \times \mathcal{F} \to [0, 1]$. Then $Q$ is said to be a **conditional probability distribution of** $X$ **given** $Y$ if

1. $Q$ is a stochastic transition kernel from $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), P_Y)$ to $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$

2. for each $A, B \in \mathcal{F}$,
$$\int_B Q(y, A) dP_Y(y) = P(X \in A, Y \in B)$$

**Note 6.2.0.3.** It is helpful to connect this notion of conditional probability with the elementary one by writing $Q(y, A) = P(X \in A|Y = y)$. If $P_Y \ll \mu$, then property (2) in the definition becomes

$$\begin{aligned}P(X \in A, Y \in B) &= \int_B Q(y, A) dP_Y(y) \\ &= \int_B P(X \in A|Y = y) f_Y(y) d\mu(y)\end{aligned}$$

as in a first course on probability.

**Exercise 6.2.0.4.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n$ and $Q : \mathbb{R}^n \times \mathcal{F} \to [0, 1]$. Suppose that for each $A \in \mathcal{F}$, $Q(\cdot, A)$ is $\mathcal{B}(\mathbb{R}^n)$-measurable, for $P_Y$-a.e. $y \in \mathbb{R}^n$, $P_{X|Y}(y, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$ and $Q(Y, A) = P(X \in A|Y)$ a.e. Then $Q$ is a conditional probability of $X$ given $Y$.

*Proof.* By assumption, for each $A \in \mathcal{F}$, $Q(\cdot, A)$ is $\mathcal{B}(\mathbb{R}^n)$-measurable and for $P_Y$-a.e. $y \in \mathbb{R}^n$, $Q(y, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$. Let $A, B \in \mathcal{F}$. Then

$$\begin{aligned}
\int_B Q(y, A) dP_Y(y) &= \int_{Y^{-1}(B)} Q(Y(\omega), A) dP(\omega) \\
&= \int_{Y^{-1}(B)} P(X \in A|Y) dP \\
&= \int_{Y^{-1}(B)} E[1_{X^{-1}(A)}|Y] dP \\
&= \int_{Y^{-1}(B)} 1_{X^{-1}(A)} dP \\
&= \int 1_{X^{-1}(A)} 1_{Y^{-1}(B)} dP \\
&= \int 1_{X^{-1}(A) \cap Y^{-1}(B)} dP \\
&= P(X \in A, Y \in B)
\end{aligned}$$

So $Q$ is a conditional probability distribution of $X$ given $Y$. $\square$

**Definition 6.2.0.5.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n$ and $\mu$ a $\sigma$-finite measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Suppose that $P_X, P_Y \ll \mu$. Then $P_{X,Y} \ll \mu^2$. Let $f_X = dP_X/d\mu$, $f_Y = dP_Y/d\mu$ and $f_{X,Y} = dP_{X,Y}/d\mu^2$. Define $f_{X|Y} : \mathbb{R}^n \times \mathbb{R}^n$ by

$$f_{X|Y}(x, y) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_Y(y)}, & y \in \operatorname{supp} Y \\ 0, & y \notin \operatorname{supp} Y \end{cases}$$

Then $f_{X|Y}$ is called the **conditional probability density of** $X$ **given** $Y$.

**Exercise 6.2.0.6.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^0_n$ and $\mu$ a $\sigma$-finite measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Suppose that $P_X, P_Y \ll \mu$. Define $Q : \mathbb{R}^n \times \mathcal{F} \to [0, 1]$ by

$$Q(y, A) = \int_A f_{X|Y}(x, y) d\mu(x)$$

Then $Q$ is a conditional probability distribution of $X$ given $Y$.

*Proof.* By the Fubini-Tonelli Theorem, for each $A \in \mathcal{F}$, $Q(\cdot, A)$ is $\mathcal{B}(\mathbb{R}^n)$-measurable and for $P_Y$-a.e. $y \in \mathbb{R}^n$, $Q(y, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$. Let $A, B \in \mathcal{F}$. Then

$$\begin{aligned}
\int_B Q(y, A) dP_Y(y) &= \int_B \left[ \int_A f_{X|Y}(x, y) d\mu(x) \right] dP_Y(y) \\
&= \int_{B \cap \operatorname{supp} Y} \left[ \int_{A \cap \operatorname{supp} Y} \frac{f_{X,Y}(x, y)}{f_Y(y)} d\mu(x) f_Y(y) \right] d\mu(y) \\
&= \int_{B \cap \operatorname{supp} Y} \left[ \int_A f_{X,Y}(x, y) d\mu(x) \right] d\mu(y) \\
&= P(X \in A, Y \in B \cap \operatorname{supp} Y) \\
&= P(X \in A, Y \in B)
\end{aligned}$$

$\square$

**Theorem 6.2.0.7.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X, Y \in L^1_n(\Omega, \mathcal{F}, P)$. Suppose that $\operatorname{Img} X \in \mathcal{B}(\mathbb{R}^n)$. Then there exists a conditional probability distribution of $Y$ given $X$.

# Chapter 7

# Markov Chains

**Definition 7.0.0.1.** Define Markov kernels, their composition and

**Definition 7.0.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space and $(X_n)_{n\in\mathbb{N}_0} \in L^0_n$. Then $(X_n)_{n\in\mathbb{N}_0}$ is said to be a **homogeneous Markov chain** if for each $A \in \mathcal{F}$ and $n \in \mathbb{N}$, $P(X_n \in A|X_1, \cdots, X_{n-1}) = P(X_1 \in A|X_0)$ a.e.

# Chapter 8

# Probabilities Induced by Isometric Group Actions

## 8.1 Applications to Bayesian Statistics

**Exercise 8.1.0.1.** Let $(\mathcal{X}, \mathcal{A})$ be a measurable space $(\Theta, d)$ a metric space, $G$ a group, $\phi : G \times \Theta \to \Theta$ an isometric group action. Suppose that $\bar{d}$ is a metric on $\Theta/G$. Let

- $H_p^\Theta$ be the Hausdorff measure on $\Theta$, $\mu_\mathcal{X}$ a measure on $\mathcal{X}$,
- $p$ a denisty on $\Theta$ and for each $\theta \in \Theta$, $p(\cdot|\theta)$ a density on $\mathcal{X}$.
- $\theta_0 \in \Theta$ and for $j \in \mathbb{N}$, $X_j \sim p(x|\theta_0)$

Suppose that $\mu_\Theta$ is $G$-invariant, $p$ is $G$-invariant and continuous on $\Theta$ and for each $x \in \mathcal{X}$, $p(x|\cdot)$ is $G$-invariant and continuous on $\Theta$. For $n \in \mathbb{N}$, set $p(\cdot|X^{(n)}) \propto f(X_1, \ldots, X_n|\cdot)p(\cdot)$. Define the posterior measure $P_{\Theta|X^{(n)}} : \mathcal{B}(\Theta) \to [0, 1]$ by

$$dP_{\Theta|X^{(n)}}(\theta) = p(\theta|X^{(n)})\, dH_p^\Theta(\theta)$$

Then there exists a density $\bar{p}(\cdot|X^{(n)})$ on $\Theta/G$ such that

$$d\bar{P}_{\Theta|X^{(n)}}(\theta) = \bar{p}(\theta|X^{(n)})\, d\bar{H}^\Theta(\theta)$$

*Proof.* Clear from previous work. $\square$

**Exercise 8.1.0.2.** Let $(\mathcal{X}, \mathcal{A})$ be a measurable space $(\Theta, d)$ a metric space, $G$ a group, $\phi : G \times \Theta \to \Theta$ an isometric group action. Suppose that $\bar{d}$ is a metric on $\Theta/G$. Let

- $H_p^\Theta$ be the Hausdorff measure on $\Theta$, $\mu_\mathcal{X}$ a measure on $\mathcal{X}$,
- $p$ a denisty on $\Theta$ and for each $\theta \in \Theta$, $p(\cdot|\theta)$ a density on $\mathcal{X}$.
- $\theta_0 \in \Theta$ and for $j \in \mathbb{N}$, $X_j \sim p(x|\theta_0)$

Suppose $p$ is $G$-invariant and continuous on $\Theta$ and for each $x \in \mathcal{X}$, $p(x|\cdot)$ is $G$-invariant and continuous on $\Theta$. For $n \in \mathbb{N}$, set $p(\cdot|X^{(n)}) \propto f(X_1, \ldots, X_n|\cdot)p(\cdot)$. Define the posterior measure $P_{\Theta|X^{(n)}} : \mathcal{B}(\Theta) \to [0, 1]$ by

$$dP_{\Theta|X^{(n)}}(\theta) = p(\theta|X^{(n)})\, dH_p^\Theta(\theta)$$

Suppose that $(P_{\Theta|X^{(n)}})_{n\in\mathbb{N}}$ concentrates on $\bar{\theta}_0 \subset \Theta$ a.s. or in probability. Then $(\bar{P}_{\Theta|X^{(n)}})_{n\in\mathbb{N}}$ concentrates a.s. or in probability on $\{\bar{\theta_0}\} \subset \Theta/G$ (i.e. is consistent a.s. or in probability)

*Proof.* Let $V \in \mathcal{N}_{\bar{\theta}_0}$. Then $\pi^{-1}(V) \in \mathcal{N}_{\bar{\theta}_0}$. By definition,

$$\begin{aligned}
\bar{P}_{\Theta|X^{(n)}}(V^c) &= P_{\Theta|X^{(n)}}(\pi^{-1}(V^c)) \\
&= P_{\Theta|X^{(n)}}(\pi^{-1}(V)^c) \\
&\xrightarrow{\text{a.s.}/P} 0
\end{aligned}$$

□

**Note 8.1.0.3.** Some examples of $G$-invariant priors would be the uniform distribution, or $N_n(0, \sigma^2 I)$ on $\mathbb{R}^n$ when acted on by $O(n)$. An example of a $G$-invariant likelihood would be $f(A|Z) \sim \text{Ber}(ZZ^T)$ as in a latent position random graph model where $Z \in \mathbb{R}^{n \times d}$ is the parameter is invariant under right multiplication by $U \in O_d$.

## Chapter 9

# Projective Systems

# Part II

# Stochastic Processes

# Chapter 10

# Basic Concepts

## 10.1 Filtrations

**Note 10.1.0.1.**
- We recall for a poset $(T, \leq)$, the Alexandrov topoplogy on $T$, denoted $\mathcal{T}_{\leq} \in \mathrm{Top}(T)$, is defined by $\mathcal{T}_{\leq} := \tau_T(\uparrow t : t \in T)$.
- We note that

**Definition 10.1.0.2.** Let $(\Omega, \mathcal{F})$, $(S, \Sigma)$ be measurable spaces and $(T, \leq_T)$ a poset.

- Let $F : T \to \downarrow(\mathcal{F}, \subset_{\sigma\text{-Alg}(\Omega)})$. Then $F$ is said to be a **filtration of** $\mathcal{F}$ if $F$ is $(\leq_T, \subset_{\sigma\text{-Alg}(\Omega)})$-monotonic.
- We define $\mathrm{Filtrn}(\mathcal{F}, T) := \{F \in \downarrow(\mathcal{F}, \subset_{\sigma\text{-Alg}(\Omega)})^T : F \text{ is monotonic}\}$.

## 10.2 Stopping Times

**Definition 10.2.0.1.** Let $(\Omega, \mathcal{F})$ be a measurable space, $T$ a poset and $\tau \in \mathrm{Meas}((\Omega, \mathcal{F}), (T, \mathcal{B}(T)))$. Then tau is said to be a **stopping time** if for each $t \in T$, $\tau^{-1}(\downarrow t) \in \mathcal{F}_t$.

**Exercise 10.2.0.2.**

## 10.3 Stochastic Processes

**Definition 10.3.0.1.** Let $(\Omega, \mathcal{F})$, $(S, \Sigma)$ be measurable spaces, $T$ a poset and $X : T \times \Omega \to S$. Then $X$ is said to be a **stochastic process** if for each $t \in T$, $X(t, \cdot) \in \mathrm{Meas}((\Omega, \mathcal{F}),\ (S, \Sigma))$.

# Chapter 11

# Martingales

# Chapter 12

# Ito Integration

## 12.1 Introduction

here are two important pages:

- almost sure math link This page is about how the covariance matrix of a random field determines the smoothness of its realizations
- https://almostsuremath.com/2010/01/03/the-stochastic-integral/ This page is about defining stochastic integration from an axiomatic approach. Lemma 3 basically defines semimartingales, but it might be better to cover rough path integration in the integration notes first and make this a special case?

If these pages do not exist in the future, try to visit a web arxiv page.

# Chapter 13

# Some Toy Models

## 13.1 Two Class Model

Goal: make simple model of resource allocation over time in the presence of two classes, class 1 and class 2, where class 1 represents labor and class 2 represents the ruling class.

- Model the population as a digraph with adjacency matrix $A(t)$ and $A(t)_{i,j}$ represents the amount of influence of vertex $i$ over vertex $j$,
- Allow for two cases (fixed classes or not) in which the class of a vertex is fixed for all time or allows for changes, try various settings for the allowed changes if not fixed
- Allow for two cases (segregation or not) in which $A(t)_{i,j} = 0$ if class of $(i, j)$ is $(1, 2)$ so that $i$ is labor class and $j$ is ruling class.
- define params like resource allocation rate $\theta(t)_k$ of the proportion of total resources allocated to class $k$, which is decided entirely by the ruling class.
- define a utility function to be maximized for each person which depends on $A$ and the parameters. For instance a person of class $k$ would prefer their $\theta(t)_k = 1$ for all time (the want all the resources all the time).
- investigate behavior under fixed and alternating class assignments, allowed or disallowed segregation, etc.
- if possible add the capabiliy for a heat-equation like interaction where the value of some diffusive quantity measured at a vertex is some parameter of a vertex considered in the utility function and is basically the average of the parameter measured at the vertices 'around it' (i guess the ones that can influence it), this would be relevant for the segregation settings
- maybe each vertex can have a happiness parameter and the happiness parameter can be the diffusive quantity, for vertex $i$ with class $k_i$, happiness should also depend on class and the happiness of others. $h(i) = f(k_i, (h(j))_{j=1}^n)$, maybe person $i$ with $k_j = 1$ influences $j$ with $k_j = 2$ even in segregation if $h(j)$ is high (because class 2 is envious of class 1's happiness, even if they have all the resources). so over time, the hapiness levels change, the adj mat changes according to happiness levels (because influence changes), etc.

## 13.2 EXTRA

**Exercise 13.2.0.1.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X$ a set $\mathcal{A}_0$ an algebra, $\mu_0 : \mathcal{A}_0 \to \mathbb{C}$ and $B : \mathcal{A}_0 \to L^2(\Omega, \mathcal{F}, P)$. Suppose that

1. $B(\varnothing) = 0$
2. for each $E, F \in \mathcal{A}_0$, if $E \cap F = \varnothing$, then $B(E \cup F) = B(E) + B(F)$
3. $E(B(E)B(F)^*) = \mu_0(E \cap F)$

Then

1. for each $E \in \mathcal{A}_0$, $\mu_0(E) = E(|B(E)|^2)$.
2. for each $E \in \mathcal{A}_0$, $0 \leq \mu_0(E) < \infty$
3. for each $E, F \in \mathcal{A}_0$, if $E \cap F = \varnothing$, then $\mu_0(E \cup F) = \mu_0(E) + \mu_0(F)$

*Proof.*

1. Clear
2. Clear
3. Let $E, F \in \mathcal{A}_0$. Suppose that $E \cap F = \varnothing$. Then

$$\begin{aligned} E(B(E)B(F)^*) &= \mu_0(E \cap F) \\ &= \mu_0(\varnothing) \\ &= E(|B(\varnothing)|^2) \\ &= E(0) \\ &= 0 \end{aligned}$$

This implies that

$$\begin{aligned} \mu_0(E \cup F) &= E(|B(E \cup F)|^2) \\ &= E(|B(E) + B(F)|^2) \\ &= E(|B(E)|^2) + E(|B(F)|^2) + 2\mathrm{Re}\, E(B(E)B(F)^*) \\ &= \mu_0(E) + \mu_0(F) + 0 \\ &= \mu_0(E) + \mu_0(F) \end{aligned}$$

□

**Definition 13.2.0.2.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X$ a set $\mathcal{A}_0$ an algebra, $\mu_0 : \mathcal{A}_0 \to [0, \infty)$ a premeasure and $B : \mathcal{A}_0 \to L^2(\Omega, \mathcal{F}, P)$. Suppose that

1. $B(\varnothing) = 0$
2. for each $E, F \in \mathcal{A}_0$, if $E \cap F = \varnothing$, then $B(E \cup F) = B(E) + B(F)$
3. $E(B(E)B(F)^*) = \mu_0(E \cap F)$

Then $B$ is said to be a **stochastic premeasure with sturcture** $\mu_0$

# Chapter 14

# TODO

Incorporate vector measures to have conditional expectations of Banach space valued functions given a $\sigma$-algebra

# Appendix A

# Summation

**Definition A.0.0.1.** Let $f : X \to [0, \infty)$, Then we define

$$\sum_{x \in X} f(x) := \sup_{\substack{F \subset X \\ F \text{ finite}}} \sum_{x \in F} f(x)$$

This definition coincides with the usual notion of summation when $X$ is countable. For $f : X \to \mathbb{C}$, we can write $f = g + ih$ where $g, h : X \to \mathbb{R}$. If

$$\sum_{x \in X} |f(x)| < \infty,$$

then the same is true for $g^+, g^-, h^+, h^-$. In this case, we may define

$$\sum_{x \in X} f(x)$$

in the obvious way.

The following note justifies the notation $\sum_{x \in X} f(x)$ where $f : X \to \mathbb{C}$.

**Note A.0.0.2.** Let $f : X \to \mathbb{C}$ and $\alpha : X \to X$ a bijection. If $\sum_{x \in X} |f(x)| < \infty$, then $\sum_{x \in X} f(\alpha(x)) = \sum_{x \in X} f(x)$.

# Appendix B

# Asymptotic Notation

**Definition B.0.0.1.** Let $X$ be a topological space, $Y, Z$ be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = o(g) \qquad \text{as } x \to x_0$$

if for each $\epsilon > 0$, there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U$,

$$\|f(x)\| \leq \epsilon \|g(x)\|$$

**Exercise B.0.0.2.** Let $X$ be a topological space, $Y, Z$ be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. If there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U \setminus \{x_0\}$, $g(x) > 0$, then

$$f = o(g) \text{ as } x \to x_0 \quad \text{iff} \quad \lim_{x \to x_0} \frac{\|f(x)\|}{\|g(x)\|} = 0$$

**Exercise B.0.0.3.** Let $X$ and $Y$ a be normed vector spaces, $A \subset X$ open and $f : A \to Y$. Suppose that $0 \in A$. If $f(h) = o(\|h\|)$ as $h \to 0$, then for each $h \in X$, $f(th) = o(|t|)$ as $t \to 0$.

*Proof.* Suppose that $f(h) = o(\|h\|)$ as $h \to 0$. Let $h \in X$ and $\epsilon > 0$. Choose $\delta' > 0$ such that for each $h' \in B(0, \delta')$, $h' \in A$ and

$$\|f(h')\| \leq \frac{\epsilon}{\|h\| + 1} \|h'\|$$

Choose $\delta > 0$ such that for each $t \in B(0, \delta)$, $th \in B(0, \delta')$. Let $t \in B(0, \delta)$. Then

$$\begin{aligned} \|f(th)\| &\leq \frac{\epsilon}{\|h\| + 1} |t| \|h\| \\ &< \epsilon |t| \end{aligned}$$

So $f(th) = o(|t|)$ as $t \to 0$. □

**Definition B.0.0.4.** Let $X$ be a topological space, $Y, Z$ be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = O(g) \qquad \text{as } x \to x_0$$

if there exists $U \in \mathcal{N}(x_0)$ and $M \geq 0$ such that for each $x \in U$,

$$\|f(x)\| \leq M \|g(x)\|$$

# Bibliography


[1] Introduction to Algebra

[2] Introduction to Analysis

[3] Introduction to Fourier Analysis

[4] Introduction to Measure and Integration